Integrated Project Control: Enhancing Cost and Schedule Performance in EPC Projects through Stochastic Earned Value Management (SEVM)

Abstract

The execution of industrial Engineering, Procurement, and Construction (EPC) projects in the 2025 era is increasingly compromised by systemic market volatility and supply chain instability. While Earned Value Management (EVM) serves as the global standard for performance monitoring, its reliance on deterministic, linear forecasting renders it inadequate for predicting outcomes in high-risk environments. This research addresses this deficiency by proposing a framework for Stochastic Earned Value Management (SEVM), designed to bridge the integration gap between cost control and risk management. By shifting from static to probabilistic modelling, the study aims to synergize quantitative risk analysis with standard EVM metrics. The methodology employs a quantitative simulation approach, utilizing Monte Carlo techniques to integrate driver-based risk distributions specifically modelled for the renewable energy sector directly into the Estimate at Completion (EAC). The expected outcome is a validated control framework that transitions project reporting from single-point estimates to probabilistic confidence levels. This research ultimately seeks to demonstrate that integrating stochastic variables into performance baselines significantly improves forecasting accuracy and decision-making resilience in complex EPC contracts

1. Introduction and Background

The Industrial Engineering, Procurement, and Construction (EPC) sector operates in an increasingly volatile environment in the 2025 era. The post-pandemic landscape has introduced systemic instability, characterized by rapid inflation, supply chain bottlenecks, and resource scarcity. In this context, the traditional "Iron Triangle" of Project Management (Cost, Time, Scope) is often compromised by external drivers that are difficult to predict using static models.

Currently, Earned Value Management (EVM) is the global standard (ANSI/EIA-748) for monitoring project performance. However, traditional EVM is deterministic; it relies on linear extrapolation of past data to predict future outcomes. This research argues that in a volatile market, deterministic methods are insufficient. To improve project resilience, there is a critical need to integrate **Risk-Based Planning** directly into performance monitoring, moving from "static" forecasting to "stochastic" (probabilistic) forecasting.

2. Problem Statement

Despite the availability of advanced management tools, EPC projects frequently suffer from cost overruns and schedule delays. A primary cause of this failure is the **"Integration Gap"** between Cost Management and Risk Management:

1. **Siloed Functions:** Risk registers (qualitative) and EVM reports (quantitative) are typically managed as separate functions. They rarely inform each other dynamically during the project lifecycle.
2. **Deterministic Bias:** Standard EVM formulas (CPI/SPI) assume that future performance will mirror past performance. They fail to account for "black swan" events or market volatility (e.g., a sudden spike in steel prices), leading to unrealistic Estimates at Completion (EAC).
3. **Management Blind Spots:** Project Managers are often presented with a single completion date or cost figure, creating a false sense of certainty that does not reflect the statistical probability of success.

3. Research Aim and Objectives

The aim of this research is to develop a **Stochastic Earned Value Management (SEVM) Framework** that integrates quantitative risk drivers into project performance forecasting.

Specific Objectives:

- To critically evaluate the limitations of traditional deterministic EVM in the context of the 2025 industrial supply chain.
- To identify and categorize key "Risk Drivers" in modern EPC projects (e.g., commodity volatility, labor availability) suitable for quantitative modeling.
- To develop a simulation-based framework that modifies standard EVM indices (CPI/SPI) using probabilistic risk distributions.
- To demonstrate the efficacy of this framework through a comparative analysis of a renewable energy case study, contrasting traditional forecasting against the proposed stochastic method.

4. Methodology

This research will adopt a **Quantitative Simulation Approach**. This method allows for the rigorous testing of management theories without the constraints of accessing live, confidential project data.

- **Phase 1: Baseline Development (The Control)** A standard Work Breakdown Structure (WBS), schedule, and budget for a representative mid-sized industrial project (e.g., a Solar PV plant) will be developed. This will serve as the "Deterministic Baseline."
- **Phase 2: Risk Quantification** Key cost and schedule elements will be assigned probability distributions (e.g., Triangular or Beta distributions) rather than fixed values, based on current industry volatility data.
- **Phase 3: Stochastic Modeling (Monte Carlo Simulation)** The research will utilize Monte Carlo simulation techniques to integrate these risk distributions into the project schedule and budget. This will generate a range of possible outcomes rather than a single fixed point.
- **Phase 4: Comparative Analysis** The forecasting accuracy of the proposed SEVM model will be compared against traditional EVM methods. The output will focus on management decision-making metrics, specifically "Confidence Levels" (e.g., P50, P80, and P90 estimates).

5. Significance of the Research

This study is significant for the field of Project Management as it addresses the modernization of Project Controls. By bridging the gap between **Risk Management** and **Earned Value Management**, this research proposes a more resilient way to manage EPC contracts. It moves the industry away from reactive "damage control" toward proactive, probabilistic planning, which is essential for navigating the economic complexities of the 2025 market.